UVM Session 1

```
package package_name_pkg;
import uvm_pkg::*;
`include "uvm_macros.svh"

class class_name extends parent_class_name;

endclass

endpackage
```

Hello World

```
module top();
 bit clk;

 initial begin
  forever
   #1 clk = ~clk;
 end

 shift_reg_if shift_regif (clk);
 shift_reg DUT(clk, shift_regif.reset, shift_regif.s

 initial begin
  run_test("shift_reg_test");
 end

endmodule
```

```
class shift_reg_env extends uvm_env;
  `uvm_component_utils(shift_reg_env)

  function new(string name = "shift_reg_env", uvm_component parent = null);
    super.new(name,parent);
  endfunction
endclass
endpackage
```

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```
interface shift_reg_if (clk);
  input clk;
  logic reset;
  logic serial_in, direction, mode;
  logic [5:0] datain, dataout;
endinterface : shift_reg_if
```

```
class shift_reg_test extends uvm_test;
  `uvm_component_utils(shift_reg_test)

  shift_reg_env env;

  function new(string name = "shift_reg_test", uvm_component parent = null);
    super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    env = shift_reg_env::type_id::create("env",this);
  endfunction

  task run_phase(uvm_phase phase);
    super.run_phase(phase);
    phase.raise_objection(this);
    #100; `uvm_info("run_phase", "Welcome to the UVM Env.", UVM_MEDIUM)
    phase.drop_objection(this);
  endtask
endclass: shift_reg_test
```

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Driving the interface

```
vlib work
vlog -f src_files.list
vsim -voptargs=+acc work.top -classdebug -uvmcontrol=all
add wave /top/shift_regif/*
run -all
```

src_files.list

File Edit View

```
shift_reg.v
shift_reg_if.sv
shift_reg_config.sv
shift_reg_driver.sv
shift_reg_env.sv
shift_reg_test.sv
top.sv
```

```
module top();
 bit clk;

 initial begin
  forever
   #1 clk = ~clk;
 end

 shift_reg_if shift_regif (clk);
 shift_reg DUT(clk, shift_regif.reset, shift_regif.serial_in, shift_regif.direction, shift_regif

 initial begin
  uvm_config_db#(virtual shift_reg_if)::set(null, "uvm_test_top", "shift_reg_IF", shift_regif);
  run_test("shift_reg_test");
 end

endmodule
```

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```
class shift_reg_config extends uvm_object;
  `uvm_object_utils(shift_reg_config)

  virtual shift_reg_if shift_reg_vif;

  function new(string name = "shift_reg_config");
    super.new(name);
  endfunction
endclass
endpackage
```

```
class shift_reg_driver extends uvm_driver;
  `uvm_component_utils(shift_reg_driver)

  virtual shift_reg_if shift_reg_vif;
  shift_reg_config shift_reg_cfg;

  function new(string name = "shift_reg_driver", uvm_component parent = null);
    super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);

    if (!uvm_config_db #(shift_reg_config)::get(this , "", "CFG", shift_reg_cfg))begin
      `uvm_fatal("build_phase" , "Driver - Unable to get configuration object")
    end
  endfunction
```

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```
  function void connect_phase(uvm_phase phase);
   super.connect_phase(phase);
   shift_reg_vif = shift_reg_cfg.shift_reg_vif;
  endfunction

  task run_phase(uvm_phase phase);
    shift_reg_vif.direction = 0;
    shift_reg_vif.mode = 0;
    shift_reg_vif.datain = 0;
    shift_reg_vif.serial_in = 0;
    shift_reg_vif.reset = 0;
    shift_reg_vif.reset = 1;
    @(negedge shift_reg_vif.clk);
    shift_reg_vif.reset = 0;
    forever begin
      @(negedge shift_reg_vif.clk);
      shift_reg_vif.direction = $random;
      shift_reg_vif.mode = $random;
      shift_reg_vif.datain = $random;
      shift_reg_vif.serial_in = $random;
      shift_reg_vif.reset = $random;
    end
  endtask

endclass
endpackage
```

```
class shift_reg_env extends uvm_env;
    `uvm_component_utils(shift_reg_env)

  shift_reg_driver driver;

  function new(string name = "shift_reg_env", uvm_component parent = null);
    super.new(name,parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    driver = shift_reg_driver::type_id::create("driver",this);
  endfunction: build_phase
endclass
```

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```
package shift_reg_test_pkg;
import shift_reg_env_pkg::*;
import shift_reg_config_pkg::*;
import uvm_pkg::*;
`include "uvm_macros.svh"

class shift_reg_test extends uvm_test;
  `uvm_component_utils(shift_reg_test)

  shift_reg_env env;
  shift_reg_config shift_reg_cfg;

  function new(string name = "shift_reg_test", uvm_component parent = null);
    super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    env = shift_reg_env::type_id::create("env",this);
    shift_reg_cfg = shift_reg_config::type_id::create("shift_reg_cfg");

    if (!uvm_config_db #(virtual shift_reg_if)::get(this, "","shift_reg_IF", shift_reg_cfg.shift_reg_vif))
      `uvm_fatal("build_phase", "Test - Unable to get the virtual interface of the shift_reg from the uvm_config_db");

    uvm_config_db #(shift_reg_config)::set(this, "*", "CFG", shift_reg_cfg);
  endfunction


  task run_phase(uvm_phase phase);
    super.run_phase(phase);
    phase.raise_objection(this);
    #100;`uvm_info("run_phase", "Welcome to the UVM Env.", UVM_MEDIUM)
    phase.drop_objection(this);
  endtask: run_phase
endclass: shift_reg_test
endpackage
```

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